

Lab 1 – Ideal Gas Law

CHE 260 – Thermodynamics and Heat Transfer

Group Members

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1 Introduction

2 Results and Discussion

2.1 Data Processing and State Selection

2.2 Part 1: Path Independence and Tank Volume Ratio

2.2.1 Processed Results: Rapid and Slow Expansion

2.2.2 Tank Volume Ratio

2.2.3 Heat Transfer and Net Heat Transfer

2.2.4 Path Independence of State Properties

2.3 Part 2: Initial Mass and Volume of the Left Tank

2.3.1 Processed Pressure, Temperature, and Mass Flow Results

2.3.2 Initial Mass and Tank Volume

2.3.3 Sources of Error and Limitations

2.3.4 Effect of Air Compressibility

3 Conclusion

References